

Name: _____

MENDELIAN GENETICS

Topic 1: Monohybrid Crosses

Genes, Alleles and Traits

Term	Definition	Example
Gene		
Allele		
Dominant		
Recessive		
Genotype		
Phenotype		

Allele Combinations

List the genotype and describe the phenotype of each of the three possible allele combinations for an autosomal trait (use “b” for the recessive allele and “B” for the dominant allele)...

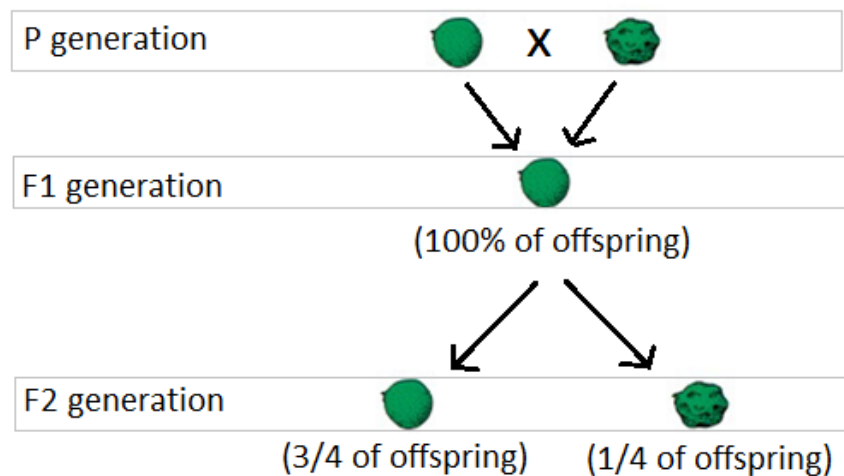
- HOMOZYGOUS DOMINANT =
- HETEROZYGOUS =
- HOMOZYGOUS RECESSIVE =

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Mendelian Genetics

The idea of dominant and recessive alleles was first described by _____ . He conducted test crosses with _____ of phenotypes and realized that for each trait, offspring displayed either the _____ or _____ phenotype rather than a _____ of the two...

- *These test crosses yielded some very interesting patterns. Explain these patterns using a Punnett Square for each cross conducted by Mendel...*



<u>P Generation</u>	<u>F1 Generation</u>

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- *Mendel's work with pea plants allowed him to develop what is known as the Law of Segregation. What is meant by this law?*

Mode of Inheritance

Because of the Law of Segregation, as long as the mode of inheritance of a particular trait and the genotypes of two parents are known, we can use a Punnett Square to determine the possible genotypes of offspring.

- AUTOSOMAL DOMINANT =

- AUTOSOMAL RECESSIVE =

Monohybrid Crosses

Example: In mice, black fur is inherited in an autosomal dominant manner over brown fur...

- List the possible genotypes of a black mouse and a brown mouse.

- If a heterozygous black mouse is crossed with a brown mouse, what is the probability that their offspring will have black fur? Brown fur?

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- c. If two heterozygous black mice are crossed, what will be the genotypic and phenotypic ratio of their offspring?
- d. Is it possible for a homozygous black mouse and a heterozygous black mouse to have “grandchildren” that are brown?

Test Crosses

- *What is the purpose of a test cross?*

- *Give an example of a test cross you could conduct to determine the genotype of an individual who demonstrates the dominant phenotype for a particular trait.*

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